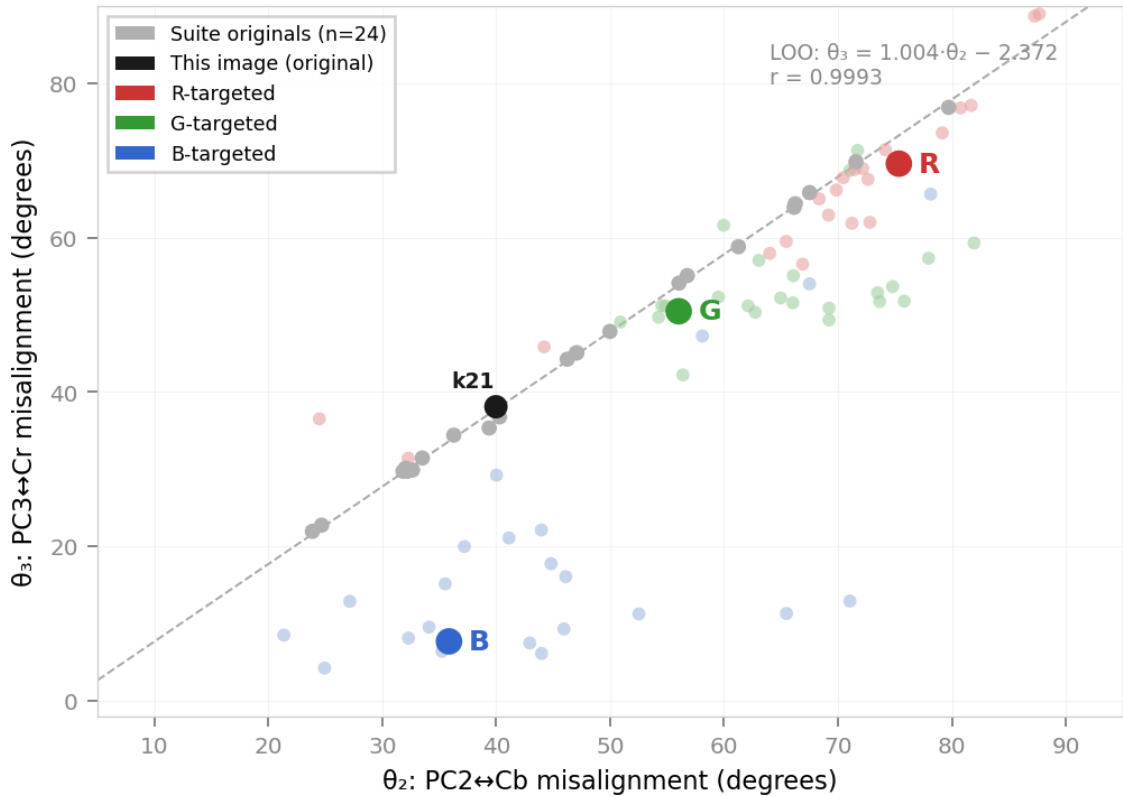


Compression Profile: KODIM21 768x512 | 24-bit RGB | PNG (lossless)

Portland Head Light | Weakly One-Dimensional (PC1 = 88.7%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8957	0.6651	0.9155	0.825	39.97°	38.1°	+0.34°
R-targeted	0.1278	0.1505	0.9747	0.418	75.35°	69.62°	-3.66°
G-targeted	0.1168	0.9129	0.1690	0.400	56.02°	50.48°	-3.39°
B-targeted	0.9495	0.0351	0.0329	0.339	35.85°	7.67°	-25.95°

R-targeted isolates red (R-G, R-B < 0.15) while preserving G-B coupling at 0.975.
G-targeted isolates green (R-G, G-B < 0.17) while preserving R-B at 0.913.
B-targeted preserves R-G at 0.950 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM21 768x512 | 24-bit RGB | PNG (lossless)

Portland Head Light | Weakly One-Dimensional (PC1 = 88.7%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

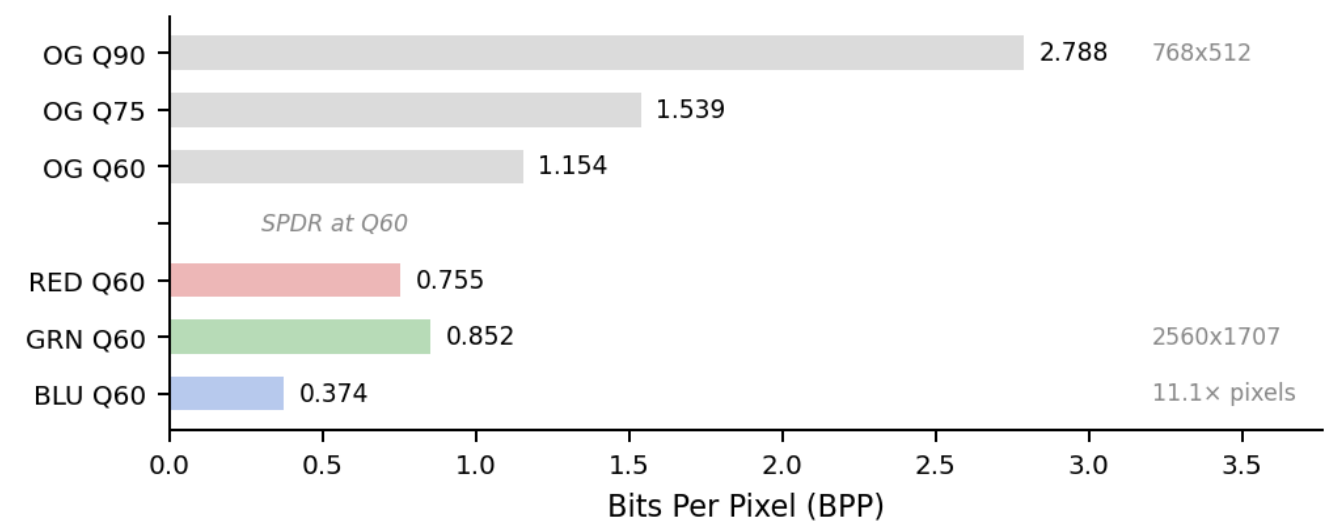
ORIGINAL (kodim21)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	622.1 KB	12.961	393,216
JPEG Q90	JPEG	768x512	133.8 KB	2.788	393,216
JPEG Q75	JPEG	768x512	73.9 KB	1.539	393,216
JPEG Q60	JPEG	768x512	55.4 KB	1.154	393,216

REDISTRIBUTED (kodim21 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	402.7 KB	0.755	4,369,920
GRN Q60	JPEG	2560x1707	454.3 KB	0.852	4,369,920
BLU Q60	JPEG	2560x1707	199.7 KB	0.374	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM21 768x512 | 24-bit RGB | PNG (lossless)

Portland Head Light | Weakly One-Dimensional (PC1 = 88.7%)

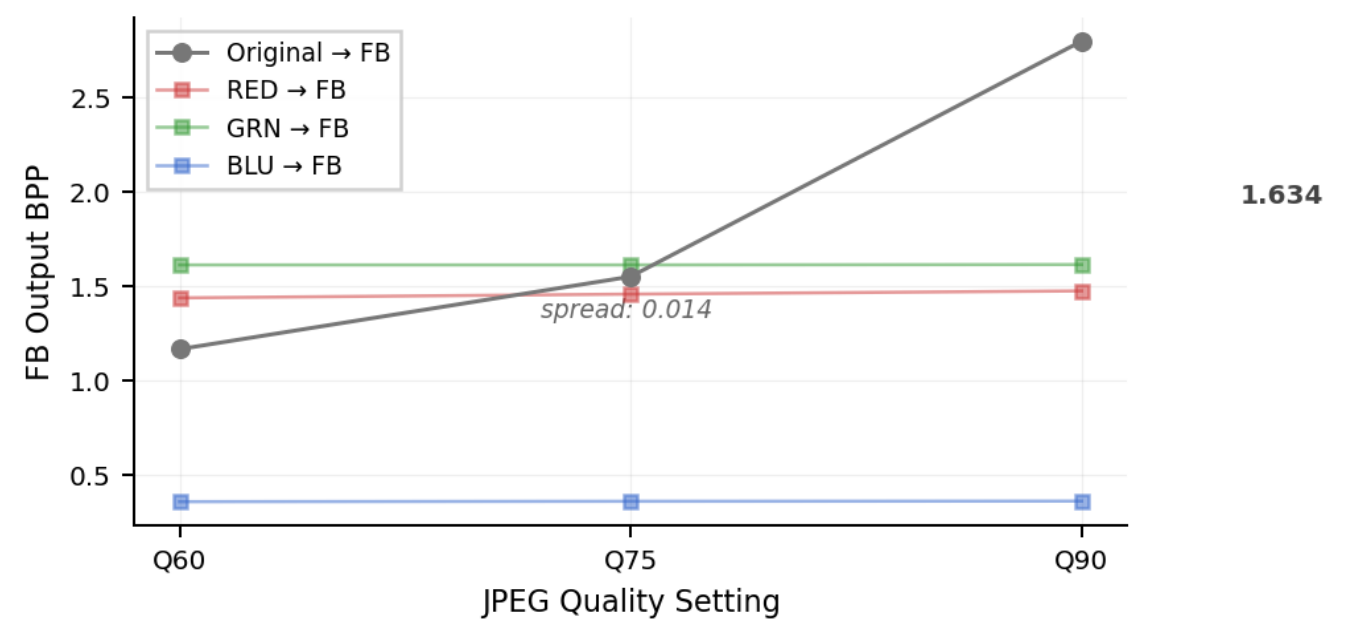
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.622	—
R-targeted → FB	2560x1707	2048x1366	1.473	43.8%
G-targeted → FB	2560x1707	2048x1366	1.613	38.5%
B-targeted → FB	2560x1707	2048x1366	0.358	86.3%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.426	45.6%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8220 vs 0.825 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.634 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.014 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim21

Best BPP (Q60, direct)	0.374	2560x1707, 4,369,920 px
Best BPP through FB	0.358	2048x1366 output
vs OG Q90 → FB	87.2%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216